

BIO101 IN-COURSE LAB PROTOCOL

Experiment: Blood Grouping (ABO and Rh Typing)

Objective: To determine the ABO and Rh blood group of a given blood sample by agglutination reaction.

Materials:

1. Blood sample
2. Glass slides
3. Lancet device
4. Mixing sticks (toothpicks or applicator sticks)
5. Serum containing **Anti-A, Anti-B, and Anti-D Antibodies**
6. Cotton swabs
7. Alcohol for disinfection
8. A marker or pencil for labeling
9. Laboratory gloves

Procedure:

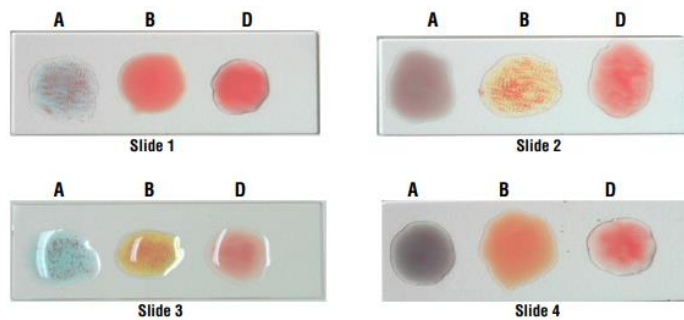


Figure SEQ Figure * ARABIC 1 Example of the Procedure

1. Divide the slide into 3 parts and label them as A, B, and D (for Anti-A, Anti-B, and Anti-D)
2. 3 drops of the blood sample are placed on the glass slide
3. Place a small drop of Anti-A serum in the area labeled Anti-A.
4. Place a small drop of Anti-B serum in the area labeled Anti-B.
5. Place a small drop of Anti-D serum in the area labeled Anti-D.
6. Use separate mixing sticks to gently mix the blood with each antibody to avoid cross-contamination.
7. Ensure the blood and antibody drops are approximately equal in volume.

Observation:

Observe for agglutination (clumping) in each section. It may occur immediately or within 1–2 minutes. Tilt the slide gently to enhance visibility of the reaction.

Interpretation of Results:

Clumping indicates the presence of the antigen; when Anti-A is added to Blood group A, there will be clumping due to the Anti-A antibody and A-Antigen reaction. But there will be no clumping between Anti-B and A-Antigen. That's how blood groups are determined. Figure 2 demonstrates the reactions observed for each blood group.

BLOOD TYPE	ANTI-A	ANTI-B	ANTI-D
O-POSITIVE			
O-NEGATIVE			
A-POSITIVE			
A-NEGATIVE			
B-POSITIVE			
B-NEGATIVE			
AB-POSITIVE			
AB-NEGATIVE			

Figure 2: Reactions between Anti-A, Anti-B, and Anti-D antibodies with each type of blood group

Table 1 provides examples of reaction types that could be observed for selected blood groups. This is not an interpretation of Fig 1.

Table 1: Example of Interpretation

Anti-A	Anti-B	Anti-D	Result
Clumping	No Clumping	Clumping	A Positive
No Clumping	Clumping	No Clumping	B Negative
Clumping	Clumping	Clumping	AB Positive
No Clumping	No Clumping	No Clumping	O Negative

- O- blood cells lack A, B, and Rh antigens. This means the recipient's immune system is unlikely to react against them, making them compatible with all blood types (A, B, AB, and O).
- AB+ plasma contains no anti-A or anti-B antibodies. This allows individuals with this type to safely receive red blood cells from any blood type.

For further information: https://youtu.be/PLhbRulwNVo?si=CwO6232X5MgIpk_y